

Overlapping insiders and method of payment in acquisitions: New tests and evidence on adverse selection

Varun Jindal*

IIMB Young Faculty Research Chair Assistant Professor of Finance
Finance and Accounting Area, Indian Institute of Management Bangalore, India

Email: varun.jindal@iimb.ac.in

ORCID ID: orcid.org/0000-0003-2915-2473

Rama Seth

Visiting Professor

Department of Finance, Copenhagen Business School, Copenhagen, Denmark

Email: rs.fi@cbs.dk

ORCID ID: orcid.org/0000-0002-7083-4102

This version: July 2020

*Corresponding author.

Acknowledgments: This paper is an adaptation of a chapter of Varun Jindal's PhD dissertation undertaken at the Indian Institute of Management Calcutta. We thank Vikas Agarwal, Shashwat Alok, Utpal Bhattacharya, Arup Ganguly, Arpita Ghosh, Woochan Kim, Jonathan Sangwook Nam, R. Narayanaswamy, Sugata Roychowdhury, Tarun Patel (discussant), Subrata Sarkar, Tracy Wang (discussant), and participants at the Australasian Finance and Banking Conference 2019 in Sydney, Australia, and the Sydney Banking and Financial Stability Conference 2019 in Sydney, Australia, for their constructive comments. We also acknowledge valuable comments from Andréanne Tremblay, who was supposed to discuss the paper at the Eastern Finance Association Annual Meeting 2020 that got cancelled eventually. The authors remain solely responsible for any remaining errors.

Overlapping insiders and method of payment in acquisitions: New tests and evidence on adverse selection

Abstract: We exploit an interesting setting of overlapping insider ownership in acquiring and target firms in India to conduct direct tests of adverse selection in stock-financed acquisitions. Using a novel dataset, we find that cash-financed deals generate greater value for acquiring firms' shareholders relative to stock-financed deals only when the insiders are non-overlapping at the acquiring and target firms. The underperformance of stock-financed deals relative to cash-financed ones in the case of non-overlapping insiders is consistent with the adverse selection effects of stock issues. Further, stock-financed deals with overlapping insiders outperform those without them. This result is consistent with the explanation that the presence of overlapping insiders on both sides of the deal mitigates adverse selection in stock-financed acquisitions due to symmetric information between the transacting parties. Our results are robust to sample selection bias and several robustness checks.

Keywords: Adverse selection; Information asymmetry; Mergers and acquisitions; Method of payment; Overlapping ownership

JEL classification: G14, G32, G34, D82

1 Introduction

Prior literature documents that method of payment in an acquisition is one of the important determinants of the acquirer's announcement period abnormal returns.^{1 2} In particular, announcements of stock-financed acquisitions are accompanied by negative returns and those of cash-financed acquisitions are associated with either positive or insignificant returns to acquiring firms' shareholders (Amihud, Lev, & Travlos, 1990; Servaes, 1991; Travlos, 1987; Wansley, Lane, & Yang, 1987).³ In other words, stock-financed acquisitions, on average, perform worse than cash-financed deals at least in the short-run.^{4 5} However, prior literature lacks conclusive evidence on why stock-financed deals underperform cash-financed ones.

The results from all prior studies about the underperformance of stock-financed acquisitions relative to cash-financed ones are based on a setting where there is inherently some information asymmetry between the acquirer and the target in an acquisition. The information asymmetry between the two parties may lead to adverse selection wherein the acquirer, which possesses better information about the true value of its stock, is likely to issue it only when it is overvalued, and consequently investors, anticipating the tendency of the acquirer's managers to

¹ See Andrade, Mitchell, and Stafford (2001), Betton, Eckbo, and Thorburn (2008), Golubov, Petmezas, and Travlos (2013), Mulherin, Netter, and Poulsen (2017) and Renneboog and Vansteenkiste (2019) for a comprehensive review of the literature.

² We use the terms 'method of payment', 'medium of exchange', and 'method of financing' interchangeably throughout the paper, and mean all the terms to imply the consideration (i.e., cash and stock) that an acquiring firm pays to target shareholders in an acquisition. Further, even though it is theoretically possible for a firm to raise cash by issuing equity and use the raised funds to pay target shareholders (Martynova & Renneboog, 2009), we do not find any evidence of acquiring firms raising any equity between the dates of announcement and those of completion of acquisitions in our sample. Thus, a cash-financed acquisition, in which an acquiring firm pays the target shareholders with cash, is financed either by internal cash reserves or a debt issue in our sample.

³ We use the terms 'acquisitions', 'takeovers', and 'mergers' synonymously throughout the paper.

⁴ Loughran and Vijh (1997) find that stock-financed acquisitions underperform cash-financed deals in the long-run as well.

⁵ Throughout the paper, we measure the performance of an acquisition based on the announcement-period wealth effects of the deal on the acquiring firm's shareholders.

issue overvalued stock, drive its share price down (Myers & Majluf, 1984).⁶ Several studies lend support to the notion that negative returns to bidders in stock-financed acquisitions are due to their overvaluation (Akbulut, 2013; Chemmanur, Paeglis, & Simonyan, 2009; Lie & Liu, 2018; Louis, 2004; Savor & Lu, 2009). Further, acquirers of both unlisted and listed targets are overvalued (Louis, 2013), indicating the presence of adverse selection in both kinds of acquirers.

In the prior settings (such as those of the US and several other markets), it has not been possible to conduct direct empirical tests on the presence of adverse selection in stock-financed deals. We exploit an exciting setting based in India where certain kinds of acquisitions have the same set of insiders on both sides of the deal (Jindal & Seth, 2019), and thus these acquisitions are likely to be free from the problem of adverse selection due to symmetric information between acquirers and targets.⁷ Thus, the Indian context allows us to examine the differential performance between deals that are likely to suffer from adverse selection and those that are not. Using a novel dataset, we investigate the following research questions in this paper.⁸ Do the returns to acquiring firms' shareholders in stock-financed acquisitions differ between when the insiders are overlapping vis-à-vis when they are not (i.e., when the adverse selection problem exists versus when it does not)? How do stock-financed acquisitions fare relative to cash-financed ones in the absence of adverse selection? We are first, to the best of our knowledge, to conduct the direct empirical tests of identifying adverse selection in stock-financed acquisitions.

⁶ Graham and Harvey (2001) also find support for the use of overvalued stock as a means to finance investments in their survey of CFOs.

⁷ We classify acquisitions in which acquiring and target firms are affiliated to the same business group as acquisitions with overlapping insiders. Since a business group, by definition, comprises of a set of firms with common ownership by insiders, intra-group acquisitions have common insider ownership in both acquiring and target firms. These kinds of acquisitions are not uncommon in markets with business groups.

⁸ The data on identifying the acquisition deals that share the common insiders in India has become available in 2014 for a period starting from 1988 onwards.

We find that when the insiders are non-overlapping on both sides of the deal, cash-financed deals generate greater value for acquiring firms' shareholders relative to stock-financed deals. The underperformance of stock-financed deals relative to cash-financed deals in the case of non-overlapping insiders is consistent with the adverse selection effects of stock issues as documented in prior literature. However, cash-financed deals could be different from stock-financed deals in several observable and non-observable aspects other than adverse selection, and these differences between the two kinds of deals could also drive the difference in their performance. To validate the presence of adverse selection in stock-financed acquisitions with non-overlapping insiders, we next focus on stock-financed deals only. Since the presence of overlapping insiders at the acquiring and target firms is likely to mitigate adverse selection in stock-financed acquisitions due to symmetric information between the transacting parties, we compare stock-financed deals that do not have overlapping insiders (i.e., deals that are likely to have the adverse selection problem) with stock-financed deals that have overlapping insiders on both sides of the deal (i.e., deals that are not likely to have the problem of adverse selection). We find that stock-financed deals with overlapping insiders outperform those without them. This result substantiates the presence of the adverse selection problem in stock-financed deals with non-overlapping insiders.

Finally, we test if adverse selection is the primary reason for a difference in performance between cash-financed and stock-financed deals. For carrying out this test, we focus on all those deals, including both cash-financed and stock-financed deals, which have overlapping insiders on both sides of the deal. If adverse selection is the primary reason for the underperformance of stock-financed deals (with non-overlapping insiders) relative to cash-financed deals, we should not find any significant difference in performance between stock-financed and cash-financed deals in the absence of the adverse selection problem. Indeed, we do not find significant differences between

the performance of stock-financed deals and that of cash-financed deals, when there is insider overlap on both sides of the deal. This result suggests that the difference between the performance of cash-financed and that of stock-financed deals is primarily due to the adverse selection effects of equity issues.

We contribute to the literature on mergers and acquisitions by empirically substantiating the presence of the adverse selection problem in stock-financed deals (with non-overlapping insiders) and also establishing that acquiring firms' shareholders do not earn significantly different returns between cash-financed and stock-financed acquisitions in the absence of the adverse selection problem. We also contribute to the otherwise scarce literature on overlapping ownership (see, for examples, Goranova, Dharwadkar, and Brandes (2010), Brooks, Chen, and Zeng (2018), and Jindal and Seth (2019)) by empirically documenting that overlapping ownership between transacting parties is beneficial in mitigating the problem of adverse selection in stock issues.

The rest of the paper is structured as follows. In Section 2, we provide a background to related literature and develop our hypotheses. Section 3 explains our research design. We describe our sample and data in Section 4 and conduct empirical analyses in Section 5. In Section 6, we conclude.

2 Background and hypotheses development

A stock-financed acquisition is a joint event representing an acquisition (i.e., an investment decision) as well as a stock issue (i.e., a financing decision) (Andrade, Mitchell, & Stafford, 2001; Rau & Stouraitis, 2011). However, the announcement effect of a stock-financed acquisition may not simply be the algebraic sum of the announcement effect of the acquisition (investment) and

that of the stock issue (financing). On the one hand, stock-financed acquisitions could be inferior investment decisions compared to cash-financed acquisitions because the use of overvalued stock as a means of payment could erode managerial discipline in undertaking prudent investments (Golubov, Petmezas, & Travlos, 2016; Jensen, 2005). On the other hand, stock-financed acquisitions could also act as a signaling mechanism to convey better growth opportunities with the acquiring firm (Jung, Kim, & Stulz, 1996; Martin, 1996). Thus, it may not be easy to know whether the lower announcement-period returns to the shareholders of acquiring firms in stock-financed acquisitions are solely due to adverse selection effects of stock issues.

The results from all prior studies about the underperformance of stock-financed acquisitions relative to cash-financed ones are based on a setting where there is inherently some information asymmetry between the acquirer and the target in an acquisition. These settings do not allow us to test the presence of adverse selection directly.⁹ Golubov, Petmezas, and Travlos (2016) make the first attempt in the literature to decompose the announcement effects of stock-financed acquisitions into those of acquisitions and stock issues separately. They estimate announcement effects of stock-issues in stock-financed acquisitions using announcement effects of seasoned equity offerings (SEOs). However, this indirect approach to estimating adverse selection effects of stock issues is laced with several issues. First, the announcement effects of SEOs are unlikely to be purely the announcement effects of stock issues. In fact, the announcement of an SEO is a joint event of the announcement of the stock issue as well as the announcement of the associated reasons for the stock issue. For instance, a firm announcing an SEO quite often also announces the purpose

⁹ While it is possible to alleviate information asymmetry to some extent in M&A deals that have common board members between the acquiring and target firms (Cai & Sevilir, 2012), the information asymmetry cannot be completely done away with in such settings, unlike that in deals than have common insiders on both sides. This is because board members, unlike insiders, are not involved in the day-to-day affairs of a firm, and they therefore do not possess complete information about the firm.

of the SEO, which could well be to finance an investment. Thus, it may not be appropriate to assume that the announcement effects of SEOs are solely due to the announcement of stock issues.¹⁰ Second, the characteristics of stock-financed acquirers and those of SEO-issuing firms could be very different.¹¹ If this possibility is true, the cross-sectional model, arrived at by using the sample of SEOs to estimate the announcement effects of stock-issues in stock-financed acquisitions, is misspecified. Finally, stock-financed acquisitions could also be different from cash-financed acquisitions, and therefore, the difference between the performance of stock-financed and that of cash-financed acquisitions may not just be due to adverse selection effects of stock issues.

In this paper, we exploit a clean and direct test of adverse selection based on the Indian context where certain types of acquisition deals have overlapping insiders on the acquirer and target sides (Jindal & Seth, 2019). We explain the implications of having overlapping insiders in acquisition deals in the following sections.

2.1 The presence of the adverse selection problem and its mitigation with overlapping insiders

When there is no overlap in insider ownership at acquiring and target firms, there exists information asymmetry between transacting parties. Taking advantage of asymmetric information, acquiring firms are likely to issue overvalued stock to target firms. Thus, we expect the adverse selection problem to exist in stock-financed acquisitions with non-overlapping insiders on two

¹⁰ Although Golubov et al. (2016) conduct two robustness checks by confining their estimation samples to SEOs which primarily raise funds for general corporate purposes and for substituting debt with equity, their results (shown in Panels D and E of Table 6 in their paper) are, however, inconsistent with their expectations across several model specifications. This poses a serious concern in drawing reliable conclusions from using the indirect approach to estimating announcement period returns due to stock issues alone using a sample of SEOs.

¹¹ Golubov et al. (2016), in fact, find that stock-financed acquirers are indeed very different from SEO-issuing firms in their sample.

sides of the deal. We should, therefore, find significantly lower returns to acquiring firms' shareholders in stock-financed acquisitions than those in cash-financed acquisitions in the absence of overlapping insiders between acquirers and targets. It allows us to state our first hypothesis (in the alternate form) below.

Hypothesis 1. *When insiders are non-overlapping between acquiring and target firms, stock-financed acquisitions underperform cash-financed ones.*

The underperformance of stock-financed acquisitions relative to cash-financed ones in the case of non-overlapping insiders, as we predict in Hypothesis 1, does not necessarily point to the presence of adverse selection in stock-financed deals. There could be many observable and non-observable differences between cash-financed and stock-financed deals, and these differences, in turn, could drive the differential returns to acquiring firms' shareholders between cash-financed and stock-financed deals. To convincingly establish the presence of adverse selection in stock-financed deals with non-overlapping insiders, we should ideally be able to compare the performance of these deals with other stock-financed deals that do not have the adverse selection problem. The presence of deals with overlapping insiders in the Indian context allows us to address this problem.

When the insiders are common on both sides of the deal, there is no information asymmetry between acquiring and target firms. In the presence of symmetric information, acquiring firms are highly unlikely to issue overvalued stock to target firms. The presence of overlapping insiders on both sides of the deal is, therefore, likely to do away with the problem of adverse selection

associated with stock issues. Thus, if stock-financed deals with non-overlapping insiders indeed suffer from adverse selection effects of stock issues, these deals should result in significantly lower returns to acquiring firms' shareholders relative to stock-financed deals that have overlapping insiders on both sides. This allows us to come up with our second hypothesis (in the alternate form).

Hypothesis 2. *When acquisitions are financed with stock, deals with overlapping insiders outperform deals that do not have overlapping insiders between acquiring and target firms.*

We represent the arguments related to the adverse selection effects of stock issues in stock-financed acquisitions in Figure 1. This figure depicts how information asymmetry in a stock-financed acquisition with non-overlapping insiders at Firm A (acquirer) and Firm B (target) may lead to the presence of adverse selection problem and consequently result in lower announcement returns to Firm A than it would in the absence of this effect. The figure also shows that the adverse selection problem is likely to be absent in another stock-financed deal with the same set of insiders at Firm A (acquirer) and Firm C (target) due to symmetric information between the two firms. Therefore, the returns to Firm A in the stock-financed acquisition of Firm C are unlikely to be affected by the adverse selection problem.

[INSERT FIGURE 1 ABOUT HERE]

2.2 Is adverse selection the primary reason for the underperformance of stock-financed acquisitions?

Even if adverse selection is present in stock-financed deals with non-overlapping insiders, we cannot attribute it as the sole reason for the lower performance of these deals relative to cash-financed ones. The observable and non-observable differences between cash-financed and stock-financed deals could partly attribute to the underperformance of stock-financed deals. To rule out the possibility that differences other than adverse selection could be the real drivers of the difference in performance between cash-financed and stock-financed deals, we should not find any significant difference between the performance of the two kinds of deals in the absence of adverse selection.

The acquisitions in which acquiring and target firms share the same set of insiders and thus have symmetric information are highly unlikely to involve the issue of overvalued stock by acquiring firms, and therefore these acquisitions are potentially free from the adverse selection problem. Thus, if adverse selection is the primary reason for the underperformance of stock-financed deals, then we should not observe any significant difference in performance between stock-financed and cash-financed deals when the adverse selection problem gets mitigated with overlapping insiders between the acquiring and target firms. Based on these arguments, we state our third hypothesis (in the null form) below.

Hypothesis 3: *When insiders are overlapping between acquiring and target firms, there is no significant difference in performance between stock-financed and cash-financed deals.*

3 Research design

To examine our research questions, we employ a linear regression model of the following form:

$$CAR_i = \beta_0 + \beta_1 INSIDER_OVERLAP_i + \beta_2 STOCK_i + \beta_3 INSIDER_OVERLAP_i * STOCK_i + \gamma' CONTROLS_i + \varepsilon_i \quad (1)$$

The dependent variable in Eq. (1), CAR_i , represents the cumulative abnormal returns over a 5-day window centered at the acquisition announcement date to the shareholders of acquirer i and computed as follows: $CAR_i(-2, +2) = \sum_{t=-2}^{+2} (R_{it} - (\alpha_i + \beta_i R_{mt}))$, where R_{mt} denotes the returns on the value-weighted NIFTY 100 index, R_{it} represents the actual or observed returns to the shareholders of acquirer i , and α_i and β_i are parameters of the market model estimated using 200 trading days long estimation window that ends 11 trading days before the announcement of the acquisition. Further, we winsorize CAR at the top as well as bottom 1% of its distribution to overcome the effect of outliers. Due to the presence of serial acquirers in our sample, we cluster standard errors at the acquirer level (Petersen, 2009). We control for acquirer, target, and deal characteristics, as documented in prior literature. In addition, we also include year fixed effects (i.e., indicators for years in which deals are announced) and industry fixed effects (i.e., indicators for acquirers' industries at the one-digit SIC level).¹² Table 1 provides the definitions of all our variables.

[INSERT TABLE 1 ABOUT HERE]

¹² Our main results remain qualitatively unchanged to employing industry fixed effects based on the Fama-French 12-industry classification.

Following the discussion leading to Hypothesis 1, we expect cash-financed deals to outperform stock-financed deals when the insiders are non-overlapping at the acquiring and target firms. This prediction empirically translates to a positive and significant coefficient on *STOCK*, i.e., a positively significant β_2 . Further, to validate the presence of adverse selection in stock-financed deals and its mitigation with overlapping insiders, we should observe, in line with Hypothesis 2, that stock-financed deals with overlapping insiders outperform those with non-overlapping insiders. This prediction translates to finding the sum of the coefficients on *INSIDER_OVERLAP* and the interaction term *INSIDER_OVERLAP* * *STOCK* to be positive and significant, i.e., a positively significant sum $\beta_1 + \beta_3$. Finally, if the difference between the performance of cash-financed and that of stock-financed deals is primarily due to adverse selection, this difference should disappear when the adverse selection problem is absent. In other words, when the insiders are overlapping, we should not find a significant difference in performance between stock-financed and cash-financed deals, which is our Hypothesis 3. To empirically validate Hypothesis 3, we should not find a significant sum of the coefficients on *STOCK* and the interaction term *INSIDER_OVERLAP* * *STOCK*, i.e., an insignificant sum $\beta_2 + \beta_3$.¹³

4 Data and descriptive statistics

We use two major sources for getting our data. We get mergers and acquisitions (M&A) data from the Thomson Reuters' Thomson One database and firm-level financial and stock market data from the Centre for Monitoring Indian Economy (CMIE)'s Prowess database.¹⁴ Further, we also

¹³ The Appendix explains how the coefficients in Eq. (1) should be interpreted.

¹⁴ CMIE Prowess data, which is sourced primarily from the annual reports of firms, has been extensively employed in literature for examining different aspects of Indian firms (see, for examples, Khanna and Palepu (2000), Bertrand,

make use of a novel dataset in CMIE Prowess, which has become available in 2014 for a period starting from 1988 onwards, to distinguish whether acquirers and targets have the set of insiders in acquisitions. Both databases have been used frequently for M&A based studies carried out in the Indian context (see, for examples, Banerjee, Banerjee, De, Jindra, and Mukhopadhyay (2014), and Jindal and Seth (2019)). In the absence of a unique and common identifier across the two databases, we employ software-based text-matching as well as hand-matching of firm names from the two databases. While matching across the two databases, we pay special attention to the changes in firm names so that deals with changes in acquirer names do not get dropped from our sample. On combining the M&A data from Thomson One with firm financial and stock data from CMIE Prowess, we obtain our initial sample.

We consider only successfully completed deals by Indian public acquirers with known deal values. We restrict our sample to the period 2000–2016 because acquisition activity was low in India prior to this period (Gubbi, Aulakh, Ray, Sarkar, & Chittoor, 2010; Jindal & Seth, 2019). Further, we keep completely cash-financed and completely stock-financed deals in our sample and exclude all such deals for which the method of payment is either unknown or hybrid. Our sample excludes deals undertaken by financial firms, financial buyers, and public sector undertakings. Following prior literature, we also exclude leveraged buyouts, spin-offs, recapitalizations, self-tenders, repurchases, privatizations, and acquisitions of remaining interest (Baker & Savaşoglu, 2002; Fich, Nguyen, & Officer, 2018; Huang, Jiang, Lie, & Yang, 2014). Finally, for our

Mehta, and Mullainathan (2002), Gopalan, Nanda, and Seru (2007), Vig (2013), Gopalan, Nanda, and Seru (2014), and Baghai and Becker (2018)). However, financial data obtained from Prowess is not exactly same as that in financial statements periodically released by firms. This discrepancy in data can be attributed to reclassification of some items by CMIE using notes to financial statement data for making the financial numbers comparable across firms (Sarkar, Sarkar, & Sen, 2008). Therefore, we validate Prowess data using data from Accord Fintech's Ace Equity database. Wherever there are substantial differences in data for our variables of interest from the two databases (i.e., numbers differing by more than 25%), we use data from annual reports of firms. Thus, we take care that our results are not merely driven by data credibility issues.

multivariate analysis, we need the data for all the required explanatory variables, and therefore, we also exclude deals with missing data for any of the explanatory variables. We get a final sample of 281 acquisition deals after applying all the sample selection filters. We detail the sample selection process in Table 2.

[INSERT TABLE 2 ABOUT HERE]

Table 3 shows the distribution of acquisition activity by year as well as by acquirer industry. We further segregate the deals into those with overlapping insiders and those that do not have overlapping insiders on both sides of the deal. Out of the total 281 deals, about 25% of the deals have overlapping insiders. We observe the highest number of deal announcements in 2015 and the lowest in 2004 (Panel A). Further, while the materials industry accounts for the largest fraction of the deal announcements, the retail industry has only one deal announcement over our sample period (Panel B).

[INSERT TABLE 3 ABOUT HERE]

Table 4 presents the descriptive statistics of the deals in our sample. As indicated by a positive mean value of our dependent variable (*CAR*), which is 1.31%, the announcements of acquisitions, on average, create wealth for the acquiring firms' shareholders in India. The positive market reaction to the announcement of deals, which is in contrast to several developed markets, has already been documented in prior literature (Banerjee et al., 2014). About 25% of the deals in our sample have been financed purely with stock, nearly 26% of the sample deals involve foreign targets, 36% of deals involve targets are in the same industry as that of acquirers, and about 36% of the acquired targets are public firms. Further, acquiring firms have concentrated ownership in the hands of their insiders, who hold about half the number of shares outstanding in these firms.

Our sample acquirers, on average, hold about 8% of the book value of their assets in the form of cash reserves, while they hold debt to the tune of 20% of their assets.

[INSERT TABLE 4 ABOUT HERE]

5 Empirical analyses

5.1 Univariate results

In Table 5, we report mean *CAR* for 3-day, 5-day, 7-day, and 9-day announcement period windows for the shareholders of acquiring firms in our sample. We observe that *CAR* for cash-financed deals, on average, is higher compared to that for stock-financed deals (though the differences are not statistically significant). However, when we segregate our sample into deals with overlapping insiders and those without them, we notice that cash-financed deals exhibit higher *CAR* relative to stock-financed deals only when the insiders are non-overlapping. On the other hand, when the insiders are overlapping, we do not observe the same trend. Overall, our univariate results indicate that cash-financed deals outperform stock-financed deals only in the presence of the adverse selection problem.

[INSERT TABLE 5 ABOUT HERE]

A closer look at stock-financed deals, as shown in Figure 2, reveals that stock-financed deals with overlapping insiders outperform those without them. This is likely indicative of the presence of the adverse selection problem in stock-financed deals with non-overlapping insiders on two sides of the deal.

[INSERT FIGURE 2 ABOUT HERE]

5.2 Multivariate results

We report the results of the multiple linear regression model based on Eq. (1) in Table 6. We employ four different model specifications. We choose *CAR* (-2,+2) as our dependent variable in each of the models. All the models have all our control variables. While Model (1) has neither the industry nor the year fixed effects, Model (4) has both kinds of fixed effects. Further, Model (2) has the acquirer industry fixed effects but not the year fixed effects, and Model (3) has the year fixed effects but not the acquirer industry fixed effects.

[INSERT TABLE 6 ABOUT HERE]

Using all the four model specifications, we observe that the coefficient on *STOCK*, β_2 , is negative and significant, and the coefficient on *INSIDER_OVERLAP*, β_1 , is not significant. A significantly negative β_2 is consistent with Hypothesis 1 that stock-financed deals underperform cash-financed deals when the insiders are non-overlapping between the acquiring and target firms. Further, the interaction term *INSIDER_OVERLAP* * *STOCK*, β_3 , obtains a positive and very significant sign. We find the sum of the coefficients $\beta_1 + \beta_3$ to be positive and significant at least at the 5% level in all our model specifications, which is consistent with Hypothesis 2, that stock-financed deals with overlapping insiders outperform stock-financed deals without overlapping insiders. The support for Hypotheses 1 and 2 substantiates the presence of the adverse selection problem in stock-financed deals that do not have overlapping insiders on both the deal sides.

Finally, we also test the significance of the sum of the coefficients, $\beta_2 + \beta_3$, and we do not find it significant. This result is consistent with Hypothesis 3 (stated in the null form) that in the presence of overlapping insiders (i.e., when the adverse selection problem is absent), the returns to acquiring firms' shareholders are not significantly different between the cash- and stock-financed acquisitions. Overall, based on the direct tests of identifying adverse selection, our results

provide convincing evidence that the lower performance of stock-financed deals relative to cash-financed ones can be attributed primarily to the adverse selection effects of stock issues.

5.3 *Sample selection bias*

Most of the corporate finance events are not random decisions (Li & Prabhala, 2007). In the context of our study, we observe the returns for only those bidders that go for acquisitions. There is a possibility that primarily those firms undertake acquisitions that expect a favorable market response to the announcements of such events (Huang et al., 2014). This possibility can potentially introduce selection bias in our sample and can contaminate our results. Following prior literature (see, for examples, Faccio and Masulis (2005) and Jindal and Seth (2019)), we employ Heckman's two-stage procedure (Heckman, 1976, 1979). The first stage models the probability of a firm making a bid as a function of its characteristics, including leverage, cash-reserves, size, working capital, and sales growth. We also include the firm's industry at the two-digit SIC level and year of estimation as other explanatory variables in the model. This step allows us to compute the inverse Mills ratio for all the observations. In the second stage, we include the inverse Mills ratio as an additional explanatory variable for our sample of acquisitions.

Table 7 reports the results of the second stage.¹⁵ We observe that the inverse Mills ratio (*INV_MILLS_RATIO*) is statistically insignificant, which indicates that we do not have selection bias in our sample. Further, our main results remain qualitatively unchanged to the inclusion of

¹⁵ The results of the first stage have not been reported for the sake of brevity. The results are available from the authors upon request.

the inverse Mills ratio, and we continue to find results consistent with all three of our hypotheses.¹⁶ Thus, we conclude that sample selection bias does not drive our results.

[INSERT TABLE 7 ABOUT HERE]

5.4 *Robustness checks*

In this section, we conduct the robustness of our main results to a variety of checks and report the results in brief in Table 8.

5.4.1 *Considering the nature of acquisitions*

An acquisition could be hostile or friendly in nature, and this could influence how the market reacts to its announcement (Schwert, 2000). Since deals with overlapping insiders are likely to be friendly acquisitions predominantly, it is possible that the friendly nature of these acquisitions drives the higher announcement effects of these deals rather than the absence of information asymmetry. Although it is rare to witness a hostile acquisition in India due to concentrated ownership of insiders in Indian firms, we still check for this possibility in our sample of deals.¹⁷ We observe that none of the deals in our sample is a hostile acquisition. Further, out of the 281 acquisitions in our sample, we observe that 271 deals are friendly deals, and the rest 10 have a neutral attitude (neither friendly nor hostile). Our results remain robust to excluding deals with neutral attitude from our analysis (Panel A of Table 8).

[INSERT TABLE 8 ABOUT HERE]

¹⁶ On including the inverse Mills ratio, however, the number of observations in our regression analyses come down because this ratio could not be calculated for some deals due to missing variables.

¹⁷ Banerjee et al. (2014) document that about 99% of the acquisitions made by firms in India are friendly acquisitions.

5.4.2 *Employing a longer event window*

For multivariate analyses, we have computed our dependent variable, *CAR*, based on the 5-day window (-2, +2) centered at the acquisition announcement date. However, our results remain qualitatively unchanged to using a slightly longer window (-3, +3) of 7 days centered at the announcement date of acquisition for computing cumulative abnormal returns (Panel B of Table 8).

5.4.3 *Employing alternative definition of industrial relatedness*

Finally, our definition of an acquisition in a related industry is based on the matching of all the four digits of the SIC code of the acquirer with that of the target. Our results continue to hold even when we match only the first digit of the SIC codes of the acquiring and target firms for defining our control variable on industry relatedness (Panel C of Table 8).

6 Discussion and conclusion

Using a novel dataset and a unique setting based on the overlapping insider ownership in acquiring and target firms in India, this paper directly tests the presence of adverse selection in stock-financed deals. Consistent with the adverse selection effects of equity issues, we first find that cash-financed deals outperform stock-financed deals when the insiders are not overlapping between the acquiring and target firms. To further substantiate the presence of adverse selection in stock-financed deals with non-overlapping insiders, we compare the performance of stock-financed deals that do not have overlapping insiders with that of ones having overlapping insiders on two sides of the deal. The presence of overlapping insiders at the acquiring and target firms is likely to mitigate adverse selection in stock-financed acquisitions due to symmetric information between the transacting firms. This allows us to directly compare the stock-financed deals that are likely to have the problem of adverse selection with ones that are unlikely to have this problem.

Consistent with the presence of the adverse selection problem in stock-financed deals with non-overlapping insiders, we show that stock-financed deals with overlapping insiders outperform those without them. Finally, when the insiders are overlapping, we do not find significant differences between the performance of stock-financed and that of cash-financed deals. This result indicates that the difference between the performance of cash-financed and that of stock-financed deals is primarily due to adverse selection effects of stock issues.

Our study has some limitations. First, given a relatively low acquisition activity by firms in India, the number of acquisitions in our sample is rather limited.¹⁸ Further, we have included only those acquisitions in our sample for which we could ascertain whether or not the acquirer and target firms share the same set of insiders. Future research can focus on a larger sample of acquisition deals to confirm the conclusions based on our study. Further, there could be a concern that our results might be affected by tunneling motives when the insiders are common on both sides of the deal. However, prior literature provides evidence that there is no significant tunneling in acquisitions in India when the acquirers and targets share the same set of insiders (Jindal & Seth, 2019). Furthermore, even if there is some tunneling taking place across deals with overlapping insiders, to the extent that tunneling is present in cash- and stock- financed deals to more or less the same extent, our results are highly unlikely to get affected by its presence. Thus, we can rule out that the possibility of tunneling in common insider deals could affect our results.

Overall, this study provides the first direct evidence on the presence of the adverse selection problem in stock-financed deals with non-overlapping insiders on the two sides. To the extent that returns to acquiring firms' shareholders in stock-financed acquisitions differ from cash-financed

¹⁸ Other M&A studies based on the Indian context also suffer from a small sample size, and their samples are limited to few hundred acquisitions. See, for examples, Banerjee et al. (2014), Bhaumik and Selarka (2012), Gubbi et al. (2010), and Jindal and Seth (2019).

deals for reasons other than the adverse selection problem, the indirect method of estimating returns purely due to stock issues based on SEOs is likely to fail. Therefore, even though some of our results corroborate those of Golubov et al. (2016), we provide the first direct and perhaps more convincing evidence on the presence of adverse selection in stock-financed acquisitions.

References

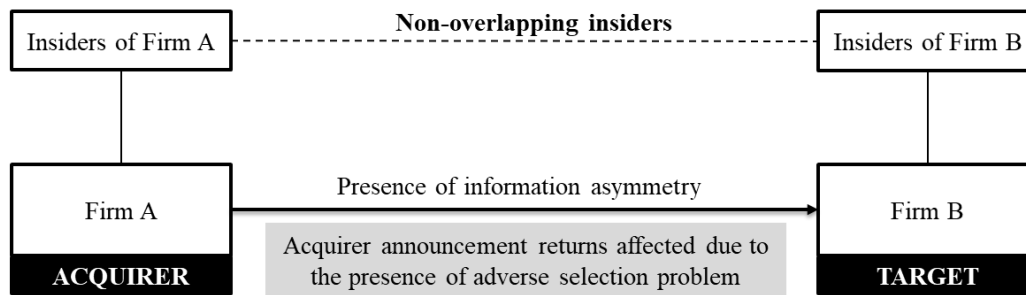
- Akbulut, M. E. (2013). Do overvaluation-driven stock acquisitions really benefit acquirer shareholders? *Journal of Financial and Quantitative Analysis*, 48(04), 1025–1055. <https://doi.org/10.1017/S0022109013000379>
- Amihud, Y., Lev, B., & Travlos, N. G. (1990). Corporate control and the choice of investment financing: The case of corporate acquisitions. *Journal of Finance*, 45(2), 603–616. <https://doi.org/10.2307/2328673>
- Andrade, G., Mitchell, M., & Stafford, E. (2001). New evidence and perspectives on mergers. *Journal of Economic Perspectives*, 15(2), 103–120. <https://doi.org/10.1257/jep.15.2.103>
- Baghai, R. P., & Becker, B. (2018). Non-rating revenue and conflicts of interest. *Journal of Financial Economics*, 127(1), 94–112. <https://doi.org/10.1016/j.jfineco.2017.10.004>
- Baker, M., & Savaşoglu, S. (2002). Limited arbitrage in mergers and acquisitions. *Journal of Financial Economics*, 64(1), 91–115. [https://doi.org/10.1016/S0304-405X\(02\)00072-7](https://doi.org/10.1016/S0304-405X(02)00072-7)
- Banerjee, P., Banerjee, P., De, S., Jindra, J., & Mukhopadhyay, J. (2014). Acquisition pricing in India during 1995–2011: Have Indian acquirers really beaten the odds? *Journal of Banking & Finance*, 38(1), 14–30. <https://doi.org/10.1016/j.jbankfin.2013.09.011>
- Bertrand, M., Mehta, P., & Mullainathan, S. (2002). Ferreting out tunneling: An application to Indian business groups. *Quarterly Journal of Economics*, 117(1), 121–148. <https://doi.org/10.1162/003355302753399463>
- Betton, S., Eckbo, B. E., & Thorburn, K. S. (2008). Corporate takeovers. In B. E. Eckbo (Ed.), *Handbook of Empirical Corporate Finance* (Vol. 2, pp. 291–429). Elsevier. <https://doi.org/10.1016/B978-0-444-53265-7.50007-X>
- Bhaumik, S. K., & Selarka, E. (2012). Does ownership concentration improve M&A outcomes in emerging markets? Evidence from India. *Journal of Corporate Finance*, 18(4), 717–726. <https://doi.org/10.1016/j.jcorpfin.2012.04.001>
- Brooks, C., Chen, Z., & Zeng, Y. (2018). Institutional cross-ownership and corporate strategy: The case of mergers and acquisitions. *Journal of Corporate Finance*, 48, 187–216. <https://doi.org/10.1016/j.jcorpfin.2017.11.003>
- Cai, Y., & Sevilir, M. (2012). Board connections and M&A transactions. *Journal of Financial Economics*, 103(2), 327–349. <https://doi.org/10.1016/j.jfineco.2011.05.017>
- Chemmanur, T. J., Paeglis, I., & Simonyan, K. (2009). The medium of exchange in acquisitions: Does the private information of both acquirer and target matter? *Journal of Corporate Finance*, 15(5), 523–542. <https://doi.org/10.1016/j.jcorpfin.2009.08.004>
- Faccio, M., & Masulis, R. W. (2005). The choice of payment method in European mergers and acquisitions. *Journal of Finance*, 60(3), 1345–1388. <https://doi.org/10.1111/j.1540-6261.2005.00764.x>
- Fich, E. M., Nguyen, T., & Officer, M. (2018). Large wealth creation in mergers and

- acquisitions. *Financial Management*, 47(4), 953–991. <https://doi.org/10.1111/fima.12212>
- Golubov, A., Petmezas, D., & Travlos, N. G. (2013). Empirical mergers and acquisitions research: A review of methods, evidence and managerial implications. In *Handbook of Research Methods and Applications in Empirical Finance* (pp. 287–313). Edward Elgar Publishing. <https://doi.org/10.4337/9780857936097.00021>
- Golubov, A., Petmezas, D., & Travlos, N. G. (2016). Do stock-financed acquisitions destroy value? New methods and evidence. *Review of Finance*, 20(1), 161–200. <https://doi.org/10.1093/rof/rfv009>
- Gopalan, R., Nanda, V., & Seru, A. (2007). Affiliated firms and financial support: Evidence from Indian business groups. *Journal of Financial Economics*, 86(3), 759–795. <https://doi.org/10.1016/j.jfineco.2006.09.008>
- Gopalan, R., Nanda, V., & Seru, A. (2014). Internal capital market and dividend policies: Evidence from business groups. *Review of Financial Studies*, 27(4), 1102–1142. <https://doi.org/10.1093/rfs/hhu004>
- Goranova, M., Dharwadkar, R., & Brandes, P. (2010). Owners on both sides of the deal: Mergers and acquisitions and overlapping institutional ownership. *Strategic Management Journal*, 31(10), 1114–1135. <https://doi.org/10.1002/smj.849>
- Graham, J. R., & Harvey, C. R. (2001). The theory and practice of corporate finance: Evidence from the field. *Journal of Financial Economics*, 60(2–3), 187–243. [https://doi.org/10.1016/S0304-405X\(01\)00044-7](https://doi.org/10.1016/S0304-405X(01)00044-7)
- Gubbi, S. R., Aulakh, P. S., Ray, S., Sarkar, M. B., & Chittoor, R. (2010). Do international acquisitions by emerging-economy firms create shareholder value? The case of Indian firms. *Journal of International Business Studies*, 41(3), 397–418. <https://doi.org/10.1057/jibs.2009.47>
- Heckman, J. J. (1976). The common structure of statistical models of truncation, sample selection and limited dependent variables and a simple estimator for such models. *Annals of Economic and Social Measurement*, 5(4), 475–492. Retrieved from <https://www.nber.org/chapters/c10491.pdf>
- Heckman, J. J. (1979). Sample selection bias as a specification error. *Econometrica*, 47(1), 153. <https://doi.org/10.2307/1912352>
- Huang, Q., Jiang, F., Lie, E., & Yang, K. (2014). The role of investment banker directors in M&A. *Journal of Financial Economics*, 112(2), 269–286. <https://doi.org/10.1016/j.jfineco.2014.02.003>
- Jensen, M. C. (2005). Agency costs of overvalued equity. *Financial Management*, 34(1), 5–19. <https://doi.org/10.1111/j.1755-053X.2005.tb00090.x>
- Jindal, V., & Seth, R. (2019). A new order of financing investments: Evidence from acquisitions by India's listed firms. *Journal of Corporate Finance*, 58, 307–328. <https://doi.org/10.1016/j.jcorpfin.2019.04.007>
- Jung, K., Kim, Y.-C., & Stulz, R. (1996). Timing, investment opportunities, managerial

- discretion, and the security issue decision. *Journal of Financial Economics*, 42(2), 159–186. [https://doi.org/10.1016/0304-405X\(96\)00881-1](https://doi.org/10.1016/0304-405X(96)00881-1)
- Khanna, T., & Palepu, K. (2000). Is group affiliation profitable in emerging markets? An analysis of diversified Indian business groups. *Journal of Finance*, 55(2), 867–891. <https://doi.org/10.1111/0022-1082.00229>
- Li, K., & Prabhala, N. R. (2007). Self-selection models in corporate finance. In *Handbook of Empirical Corporate Finance* (Vol. 2, pp. 37–86). Elsevier. <https://doi.org/10.1016/B978-0-444-53265-7.50016-0>
- Lie, E., & Liu, Y. (2018). Corporate cash holdings and acquisitions. *Financial Management*, 47(1), 159–173. <https://doi.org/10.1111/fima.12185>
- Loughran, T., & Vijh, A. M. (1997). Do long-term shareholders benefit from corporate acquisitions? *Journal of Finance*, 52(5), 1765–1790. <https://doi.org/10.1111/j.1540-6261.1997.tb02741.x>
- Louis, H. (2004). Earnings management and the market performance of acquiring firms. *Journal of Financial Economics*, 74(1), 121–148. <https://doi.org/10.1016/j.jfineco.2003.08.004>
- Louis, H. (2013). Are stock-for-stock acquirers of unlisted targets really less overvalued? *Financial Management*, 42(4), 901–929. <https://doi.org/10.1111/fima.12022>
- Martin, K. J. (1996). The method of payment in corporate acquisitions, investment opportunities, and management ownership. *Journal of Finance*, 51(4), 1227–1246. <https://doi.org/10.2307/2329393>
- Martynova, M., & Renneboog, L. (2009). What determines the financing decision in corporate takeovers: Cost of capital, agency problems, or the means of payment? *Journal of Corporate Finance*, 15(3), 290–315. <https://doi.org/10.1016/j.jcorpfin.2008.12.004>
- Mulherin, J. H., Netter, J. M., & Poulsen, A. B. (2017). The evidence on mergers and acquisitions: A historical and modern report. In *Handbook of the Economics of Corporate Governance* (1st ed., pp. 235–290). Elsevier B.V. <https://doi.org/10.1016/bs.hecg.2017.11.006>
- Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, 13(2), 187–221. [https://doi.org/10.1016/0304-405X\(84\)90023-0](https://doi.org/10.1016/0304-405X(84)90023-0)
- Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1), 435–480. <https://doi.org/10.1093/rfs/hhn053>
- Rau, P. R., & Stouraitis, A. (2011). Patterns in the timing of corporate event waves. *Journal of Financial and Quantitative Analysis*, 46(1), 209–246. <https://doi.org/10.1017/S0022109010000694>
- Renneboog, L., & Vansteenkiste, C. (2019). Failure and success in mergers and acquisitions. *Journal of Corporate Finance*, 58, 650–699. <https://doi.org/10.1016/j.jcorpfin.2019.07.010>

- Sarkar, J., Sarkar, S., & Sen, K. (2008). Board of directors and opportunistic earnings management: Evidence from India. *Journal of Accounting, Auditing & Finance*, 23(4), 517–551. <https://doi.org/10.1177/0148558X0802300405>
- Savor, P. G., & Lu, Q. (2009). Do stock mergers create value for acquirers? *Journal of Finance*, 64(3), 1061–1097. <https://doi.org/10.1111/j.1540-6261.2009.01459.x>
- Schwert, G. W. (2000). Hostility in takeovers: In the eyes of the beholder? *Journal of Finance*, 55(6), 2599–2640. <https://doi.org/10.1111/0022-1082.00301>
- Servaes, H. (1991). Tobin's Q and the gains from takeovers. *Journal of Finance*, 46(1), 409–419. <https://doi.org/10.1111/j.1540-6261.1991.tb03758.x>
- Travlos, N. G. (1987). Corporate takeover bids, methods of payment, and bidding firms' stock returns. *Journal of Finance*, 42(4), 943. <https://doi.org/10.2307/2328300>
- Vig, V. (2013). Access to collateral and corporate debt structure: Evidence from a natural experiment. *Journal of Finance*, 68(3), 881–928. <https://doi.org/10.1111/jofi.12020>
- Wansley, J. W., Lane, W. R., & Yang, H. C. (1987). Gains to bidder firms in cash and securities transactions. *Financial Review*, 22(4), 403–414. <https://doi.org/10.1111/j.1540-6288.1987.tb01263.x>

a) Stock-financed acquisition with non-overlapping insiders on the acquiring and target sides



b) Stock-financed acquisition with overlapping insiders on the acquiring and target sides

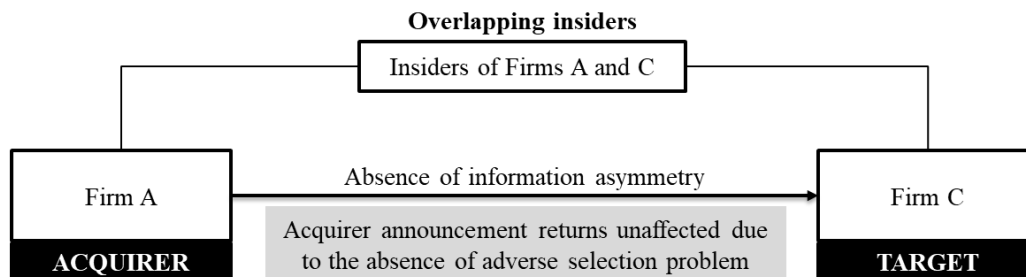


Figure 1: Classification of stock-financed acquisitions into deals with overlapping and non-overlapping insiders

This figure presents the classification of stock-financed acquisitions into two broad categories: deals with non-overlapping insiders and deals with overlapping insiders on the acquiring and target sides. While a stock-financed acquisition with non-overlapping insiders on two sides of the deal is likely to suffer from the adverse selection effect associated with the issuance of stock, a deal that has the same set of insiders between the acquirer and the target is likely to be free from the adverse selection problem.

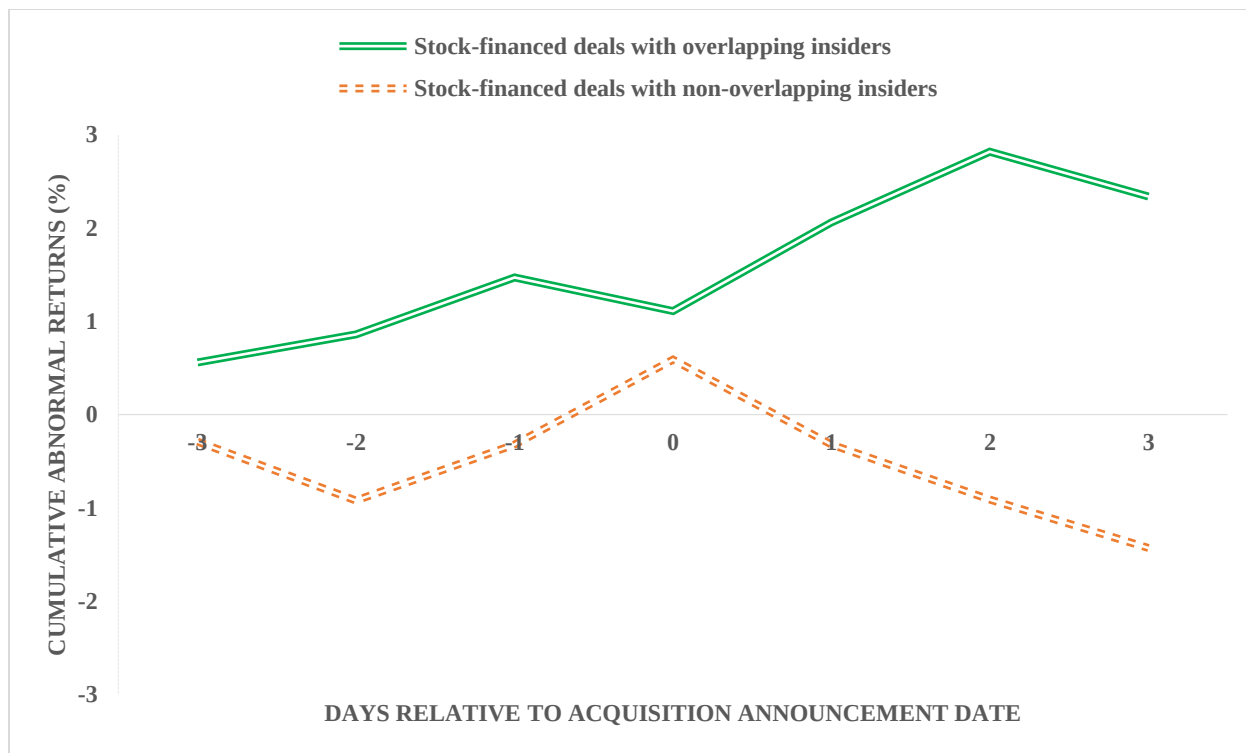


Figure 2: Market reaction to the announcement of stock-financed acquisitions

This figure shows the performance of stock-financed acquisitions separately for those with overlapping insiders and those without them and as measured by the average cumulative abnormal returns to acquiring firms' shareholders around the announcement of these deals. The deal announcement date is centered at day '0'. Cumulative abnormal returns start three days before the deal announcement date and go up to three days after the announcement date.

Table 1: Variable definitions and data sources

Variable	Definition	Source
CAR	Cumulative abnormal returns to the acquiring firm's shareholders centered at the acquisition announcement date and computed using the market model with its parameters estimated using a 200-trading-day estimation window ending 11 trading days prior to the acquisition announcement date	ThomsonOne, CMIE Prowess
CASH_TO_ASSETS	Ratio of cash and cash equivalents to total assets of the acquirer at the end of the financial year immediately preceding the acquisition announcement	CMIE Prowess
CROSS_BORDER	Equal to 1 if the target is not based in India, and zero otherwise	ThomsonOne
DEBT_TO_ASSETS	Ratio of debt to total assets of the acquirer at the end of the financial year immediately preceding the acquisition announcement	CMIE Prowess
IND_REL	Equal to 1 if the acquirer and the target share the same four-digit SIC code, and zero otherwise	ThomsonOne
INSIDER_OVERLAP	Equal to 1 if the acquirer is a group-affiliated firm and it acquires another firm from the same group, and zero otherwise	CMIE Prowess
INSIDER_OWEN	Proportion of the total shares held by the promoter group (including individuals as well as corporate bodies acting as promoters) of the acquirer at the end of the quarter immediately preceding the acquisition announcement	CMIE Prowess
INSIDER_OWEN_SQ	Square of the proportion of total shares held by the promoter group (including individuals as well as corporate bodies acting as promoters) of the acquirer at the end of the quarter immediately preceding the acquisition announcement	CMIE Prowess
MARKET_TO_BOOK	Sum of the acquirer's market value of equity and book value of debt divided by its total assets at the end of the financial year immediately preceding the acquisition announcement	CMIE Prowess
REL_SIZE	Size of the deal relative to the size of the acquirer, arrived at by dividing the deal size (converted to Indian Rupees using USD-to-Rupee Exchange Rate) with the market capitalization of the acquirer at the end of the financial year immediately preceding the acquisition announcement	ThomsonOne, RBI, CMIE Prowess
STOCK	Equal to 1 if the acquirer pays the target shareholders with its stock, and zero otherwise	ThomsonOne
TARGET_PUBLIC	Equal to 1 if the target is a publicly listed firm, and zero otherwise	ThomsonOne
TOTAL_ASSETS	Natural logarithm of the total assets of the acquirer at the end of the financial year immediately preceding the acquisition announcement	CMIE Prowess

Table 2: Sample selection

This table details the step-by-step process to arrive at the final sample of 281 acquisitions during the period 2000–2016.

Step	Count
Number of acquisitions announced and completed by Indian public acquirers between 2000 and 2016 with known deal values	1,472
<i>Less:</i> deals with an unknown or undisclosed or hybrid method of payment	(754)
<i>Less:</i> deals made by acquirers which could not be matched with CMIE Prowess	(12)
<i>Less:</i> deals made by public sector undertakings	(19)
<i>Less:</i> deals made by financial firms and financial buyers	(93)
<i>Less:</i> leveraged buyouts, spin-offs, recapitalizations, self-tenders, repurchases, privatizations, and acquisitions of remaining interest	(111)
<i>Less:</i> deals where it could not be ascertained whether there are overlapping insiders on both sides of the deal or not	(143)
<i>Less:</i> reduction in the number of observations due to clubbing of deals with same announcement dates, acquirers and targets	(15)
<i>Less:</i> deals where data on any of the explanatory variables is missing	(44)
Final Sample	281

Table 3: Acquisition activity

This table lists the distribution of acquisition deals in our sample during the period 2000–2016. Panel A reports the number, as well as the percentage, of acquisitions made by acquirers in India across each year. Panel B provides the distribution of acquisitions based on the acquirers' broad industry classification.

Panel A: Distribution of the number of acquisitions in the sample by year

	Overall		Deals with overlapping insiders		Deals without overlapping insiders	
	Number	Percentage	Number	Percentage	Number	Percentage
2000	9	3.2	4	5.7	5	2.4
2001	6	2.1	2	2.9	4	1.9
2002	10	3.6	2	2.9	8	3.8
2003	12	4.3	1	1.4	11	5.2
2004	3	1.1	2	2.9	1	0.5
2005	15	5.3	7	10.0	8	3.8
2006	17	6.1	2	2.9	15	7.1
2007	27	9.6	4	5.7	23	10.9
2008	17	6.1	5	7.1	12	5.7
2009	22	7.8	11	15.7	11	5.2
2010	26	9.3	8	11.4	18	8.5
2011	18	6.4	4	5.7	14	6.6
2012	20	7.1	10	14.3	10	4.7
2013	14	5.0	2	2.9	12	5.7
2014	9	3.2	0	0.0	9	4.3
2015	37	13.2	5	7.1	32	15.2
2016	19	6.8	1	1.4	18	8.5
Total	281	100.0	70	100.0	211	100.0
Percentage	100.0		24.9		75.1	

Panel B: Distribution of the number of acquisitions in the sample by acquirer industry

	Overall		Deals with overlapping insiders		Deals without overlapping insiders	
	Number	Percentage	Number	Percentage	Number	Percentage
Consumer Products and Services	20	7.1	1	1.4	19	9.0
Consumer Staples	30	10.7	12	17.1	18	8.5
Energy and Power	15	5.3	6	8.6	9	4.3
Healthcare	35	12.5	2	2.9	33	15.6
High Technology	61	21.7	8	11.4	53	25.1
Industrials	29	10.3	8	11.4	21	10.0
Materials	66	23.5	29	41.4	37	17.5
Media and Entertainment	10	3.6	2	2.9	8	3.8
Real Estate	8	2.9	0	0.0	8	3.8
Retail	1	0.4	1	1.4	0	0.0
Telecommunications	6	2.1	1	1.4	5	2.4
Total	281	100.0	70	100.0	211	100.0
Percentage	100.0		24.9		75.1	

Table 4: Descriptive statistics

The table presents the descriptive statistics of the acquisition deals in our sample during the period 2000–2016. The definitions of variables are provided in Table 1.

Variable	N	Mean	St Dev	Min	Q1	Median	Q3	Max
CAR	281	1.31	7.80	-16.48	-3.52	0.70	4.83	27.32
INSIDER_OVERLAP	281	0.25	0.43	0	0	0	0	1
STOCK	281	0.25	0.43	0	0	0	0	1
CROSS_BORDER	281	0.26	0.44	0	0	0	1	1
REL_SIZE	281	0.57	2.98	0.00	0.01	0.07	0.21	42.82
IND_REL	281	0.36	0.48	0	0	0	1	1
CASH_TO_ASSETS	281	0.08	0.12	-0.05	0.01	0.03	0.11	0.72
DEBT_TO_ASSETS	281	0.20	0.17	0.00	0.06	0.17	0.30	0.73
TOTAL_ASSETS	281	9.52	1.93	4.03	8.19	9.45	10.85	14.74
INSIDER_OWN	281	0.50	0.19	0.08	0.31	0.51	0.64	0.88
INSIDER_OWN_SQ	281	0.28	0.20	0.01	0.10	0.26	0.41	0.78
MARKET_TO_BOOK	281	2.17	2.04	0.10	0.82	1.60	2.72	14.41
TARGET_PUBLIC	281	0.36	0.48	0	0	0	1	1

Table 5: Univariate results

This table reports the n -day cumulative abnormal returns centered at the acquisition announcement date to the shareholders of acquirer i and computed as follows: $CAR_i(-n, +n) = \sum_{t=-n}^{+n} (R_{it} - (\alpha_i + \beta_i R_{mt}))$, where R_{mt} denotes the returns on the value-weighted NIFTY 100 index, R_{it} represents the observed returns to the shareholders of acquirer i , and α_i and β_i are parameters of the market model estimated using 200 trading days long estimation window that ends 11 trading days before the acquisition of the acquisition. $CAR_i(-n, +n)$ has been winsorized at the top and bottom 1% of its distribution to mitigate the effect of outliers. The p -values are provided in parentheses. ***, **, and * denote statistical significance based on two-tailed tests at the 1%, 5%, and 10% levels, respectively.

	Overall (N = 281)			Deals with overlapping insiders (N = 70)			Deals without overlapping insiders (N = 211)		
	Cash (N = 211)	Stock (N = 70)	Difference (Cash - Stock)	Cash (N = 32)	Stock (N = 38)	Difference (Cash - Stock)	Cash (N = 179)	Stock (N = 32)	Difference (Cash - Stock)
CAR (-1,+1)	1.39*** (0.002)	0.76 (0.377)	0.64 (0.478)	0.44 (0.570)	0.89 (0.475)	-0.45 (0.768)	1.57*** (0.002)	0.60 (0.613)	0.97 (0.445)
CAR (-2,+2)	1.45*** (0.004)	0.88 (0.438)	0.57 (0.596)	-0.61 (0.495)	2.03 (0.197)	-2.63 (0.162)	1.82*** (0.001)	-0.48 (0.772)	2.30 (0.125)
CAR (-3,+3)	1.77*** (0.002)	0.45 (0.726)	1.31 (0.291)	0.28 (0.816)	1.98 (0.230)	-1.70 (0.415)	2.03*** (0.002)	-1.36 (0.513)	3.39* (0.053)
CAR (-4,+4)	1.66** (0.011)	0.12 (0.931)	1.53 (0.271)	0.57 (0.715)	0.80 (0.670)	-0.23 (0.926)	1.85*** (0.010)	-0.68 (0.762)	2.53 (0.191)

Table 6: Multivariate results of market reactions to bidders' stocks for acquisition announcements

This table reports the results for a set of OLS regression models of the determinants of stock market reactions to bidders' stocks around the announcement of acquisitions. The dependent variable, CAR_i , represents the 5-day cumulative abnormal returns centered at the acquisition announcement date to the shareholders of acquirer i and computed as follows: $CAR_i(-2, +2) = \sum_{t=-2}^{+2} (R_{it} - (\alpha_i + \beta_i R_{mt}))$, where R_{mt} denotes the returns on the value-weighted NIFTY 100 index, R_{it} represents the observed returns to the shareholders of acquirer i , and α_i and β_i are parameters of the market model estimated using 200 trading days long estimation window that ends 11 trading days before the acquisition of the acquisition. $CAR_i(-2, +2)$ has been winsorized at the top and bottom 1% of its distribution to mitigate the effect of outliers. The p -values are provided in parentheses and are based on heteroskedasticity-robust standard errors clustered by acquirer. The definitions of the variables are provided in Table 1. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	CAR (-2,+2)			
	(1)	(2)	(3)	(4)
INSIDER_OVERLAP (β_1)	-0.357 (0.757)	-0.852 (0.460)	-0.253 (0.845)	-1.075 (0.417)
STOCK (β_2)	-3.298* (0.058)	-3.954** (0.030)	-3.516** (0.049)	-4.594** (0.017)
INSIDER_OVERLAP * STOCK (β_3)	5.416** (0.030)	6.565*** (0.010)	5.865** (0.035)	7.641*** (0.006)
CROSS_BORDER	2.400* (0.060)	2.292* (0.089)	2.342* (0.065)	2.012 (0.147)
REL_SIZE	0.396** (0.025)	0.410** (0.022)	0.415** (0.022)	0.435** (0.020)
IND_REL	1.049 (0.313)	0.934 (0.368)	1.611 (0.125)	1.614 (0.129)
CASH_TO_ASSETS	8.442** (0.037)	8.885** (0.035)	7.948* (0.061)	7.847* (0.076)
DEBT_TO_ASSETS	1.218 (0.715)	-0.203 (0.948)	0.053 (0.988)	-1.858 (0.551)
TOTAL_ASSETS	-0.647** (0.027)	-0.517* (0.086)	-0.605* (0.071)	-0.377 (0.256)
INSIDER_OWN	-8.780 (0.465)	-6.502 (0.592)	-4.487 (0.710)	-1.480 (0.900)
INSIDER_OWN_SQ	10.794 (0.362)	8.166 (0.493)	5.531 (0.653)	1.731 (0.886)
MARKET_TO_BOOK	-0.182 (0.380)	-0.238 (0.310)	-0.251 (0.258)	-0.336 (0.173)
TARGET_PUBLIC	-0.174 (0.869)	-0.016 (0.987)	-0.609 (0.591)	-0.559 (0.621)
Intercept (β_0)	7.215 (0.117)	2.710 (0.584)	7.229 (0.250)	1.473 (0.814)

Acquirer industry fixed effects	No	Yes	No	Yes
Year fixed effects	No	No	Yes	Yes
Clustered standard errors	Yes	Yes	Yes	Yes
Observations	281	281	281	281
Adj. R ²	0.061	0.070	0.053	0.071
$\beta_1 + \beta_3 = 0$	5.059** (0.024)	5.713*** (0.010)	5.612** (0.020)	6.566*** (0.005)
$\beta_2 + \beta_3 = 0$	2.118 (0.241)	2.611 (0.150)	2.349 (0.312)	3.047 (0.173)

Table 7: Addressing sample selection bias in multivariate results of market reactions to bidders' stocks for acquisition announcements

This table reports the results for a set of OLS regression models of the determinants of stock market reactions to bidders' stocks around the announcement of acquisitions. The dependent variable, CAR_i , represents the 5-day cumulative abnormal returns centered at the acquisition announcement date to the shareholders of acquirer i and computed as follows: $CAR_i(-2, +2) = \sum_{t=-2}^{+2} (R_{it} - (\alpha_i + \beta_i R_{mt}))$, where R_{mt} denotes the returns on the value-weighted NIFTY 100 index, R_{it} represents the observed returns to the shareholders of acquirer i , and α_i and β_i are parameters of the market model estimated using 200 trading days long estimation window that ends 11 trading days before the acquisition of the acquisition. INV_MILLS_RATIO denotes the inverse Mills ratio calculated using a probit model that predicts the probability of a firm making an acquisition, and it takes care of the selection bias in our sample. $CAR_i(-2, +2)$ has been winsorized at the top and bottom 1% of its distribution to mitigate the effect of outliers. The p -values are provided in parentheses and are based on heteroskedasticity-robust standard errors clustered by acquirer. The definitions of the variables are provided in Table 1. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	CAR (-2,+2)
INSIDER_OVERLAP (β_1)	-1.077 (0.402)
STOCK (β_2)	-3.632* (0.058)
INSIDER_OVERLAP * STOCK (β_3)	6.196** (0.031)
CROSS_BORDER	1.941 (0.204)
REL_SIZE	0.363* (0.053)
IND_REL	1.496 (0.193)
CASH_TO_ASSETS	8.811* (0.057)
DEBT_TO_ASSETS	-3.591 (0.346)
TOTAL_ASSETS	-0.137 (0.863)
INSIDER_OWN	3.829 (0.763)
INSIDER_OWN_SQ	-3.825 (0.771)
MARKET_TO_BOOK	-0.239 (0.385)
TARGET_PUBLIC	-0.360 (0.761)
INV_MILLS_RATIO	1.842 (0.604)

Intercept (β_0)	-5.636 (0.720)
Acquirer industry fixed effects	Yes
Year fixed effects	Yes
Clustered standard errors	Yes
Observations	266
Adj. R^2	0.041
$\beta_1 + \beta_3 = 0$	5.119** (0.039)
$\beta_2 + \beta_3 = 0$	2.564 (0.281)

Table 8: Robustness checks

This table reports the results of robustness checks for a set of OLS regression models of the determinants of stock market reactions to bidders' stocks around the announcement of acquisitions. The dependent variable, CAR_i , represents the n -day cumulative abnormal returns centered at the acquisition announcement date to the shareholders of acquirer i and computed as follows: $CAR_i(-n, +n) = \sum_{t=-n}^{+n} (R_{it} - (\alpha_i + \beta_i R_{mt}))$, where R_{mt} denotes the returns on the value-weighted NIFTY 100 index, R_{it} represents the observed returns to the shareholders of acquirer i , and α_i and β_i are parameters of the market model estimated using 200 trading days long estimation window that ends 11 trading days before the acquisition of the acquisition. INV_MILLS_RATIO denotes the inverse Mills ratio calculated using a probit model that predicts the probability of a firm making an acquisition, and it takes care of the selection bias in our sample. $CAR_i(-n, +n)$ has been winsorized at the top and bottom 1% of its distribution to mitigate the effect of outliers. The p -values are provided in parentheses and are based on heteroskedasticity-robust standard errors clustered by acquirer. The definitions of the variables are provided in Table 1. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Restricting the sample to friendly deals

	CAR (-2,+2)	
	(1)	(2)
INSIDER_OVERLAP (β_1)	-1.150 (0.403)	-1.128 (0.397)
STOCK (β_2)	-4.939** (0.011)	-3.963** (0.040)
INSIDER_OVERLAP * STOCK (β_3)	7.835*** (0.004)	6.368** (0.024)
INV_MILLS_RATIO		2.521 (0.484)
Control variables	Yes	Yes
Acquirer industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Clustered standard errors	Yes	Yes
Observations	271	256
Adj. R ²	0.084	0.056
$\beta_1 + \beta_3 = 0$	6.685*** (0.004)	5.240** (0.032)
$\beta_2 + \beta_3 = 0$	2.896 (0.187)	2.405 (0.300)

Panel B: Employing CAR (-3, +3) as the dependent variable

	CAR (-3,+3)	
	(1)	(2)
INSIDER_OVERLAP (β_1)	-0.601 (0.690)	-0.440 (0.770)
STOCK (β_2)	-6.001** (0.010)	-4.860** (0.044)
INSIDER_OVERLAP * STOCK (β_3)	7.764** (0.011)	6.616** (0.040)
INV_MILLS_RATIO		3.936 (0.360)
Control variables	Yes	Yes
Acquirer industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Clustered standard errors	Yes	Yes
Observations	281	266
Adj. R ²	0.042	0.027
$\beta_1 + \beta_3 = 0$	7.163*** (0.006)	6.176** (0.026)
$\beta_2 + \beta_3 = 0$	1.763 (0.468)	1.756 (0.496)

Panel C: Employing alternative definition of industry relatedness

	CAR (-2,+2)	
	(1)	(2)
INSIDER_OVERLAP (β_1)	-1.313 (0.322)	-1.292 (0.314)
STOCK (β_2)	-4.469** (0.022)	-3.492* (0.076)
INSIDER_OVERLAP * STOCK (β_3)	7.765*** (0.006)	6.358** (0.030)
INV_MILLS_RATIO		1.983 (0.576)
Control variables	Yes	Yes
Acquirer industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Clustered standard errors	Yes	Yes
Observations	281	266
Adj. R ²	0.074	0.041
$\beta_1 + \beta_3 = 0$	6.452*** (0.006)	5.066** (0.044)
$\beta_2 + \beta_3 = 0$	3.296 (0.138)	2.866 (0.228)

Appendix: The interpretation of coefficients

This table captures how the coefficients in the following model (regression Eq. (1)) should be interpreted: $CAR_i = \beta_0 + \beta_1 INSIDER_OVERLAP_i + \beta_2 STOCK_i + \beta_3 INSIDER_OVERLAP_i * STOCK_i + \gamma' CONTROLS_i + \varepsilon_i$. The definitions of variables are provided in Table 1. Panel A shows which coefficients in Eq. (1) capture the announcements effects of two different types of deals, cash-financed and stock-financed deals, depending on whether the insiders are overlapping or not between the acquiring and target firms in an acquisition. Panel B provides the interpretation of the coefficients that compare the announcement effects of different kinds of deals.

Panel A: Coefficients representing the announcement effects of different kinds of deals

Deal	Insiders	Coefficient	
Cash-financed	Non-overlapping	β_0	[1]
Cash-financed	Overlapping	$\beta_0 + \beta_1$	[2]
Stock-financed	Non-overlapping	$\beta_0 + \beta_2$	[3]
Stock-financed	Overlapping	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	[4]

Panel B: Interpretation of coefficients for comparing announcement effects of different kinds of deals

Coefficient		Interpretation	Expected sign
β_1	[2] – [1]	CAR of deals with overlapping insiders relative to that of deals with non-overlapping insiders when deals are financed with cash	Not significant
β_2	[3] – [1]	CAR of stock-financed deals relative to that of cash-financed deals when insiders are non-overlapping	Negative [Hypothesis 1]
$\beta_1 + \beta_3$	[4] – [3]	CAR of deals with overlapping insiders relative to that of deals with non-overlapping insiders when deals are financed with stock	Positive [Hypothesis 2]
$\beta_2 + \beta_3$	[4] – [2]	CAR of stock-financed deals relative to that of cash-financed deals when insiders are overlapping	Not significant [Hypothesis 3]